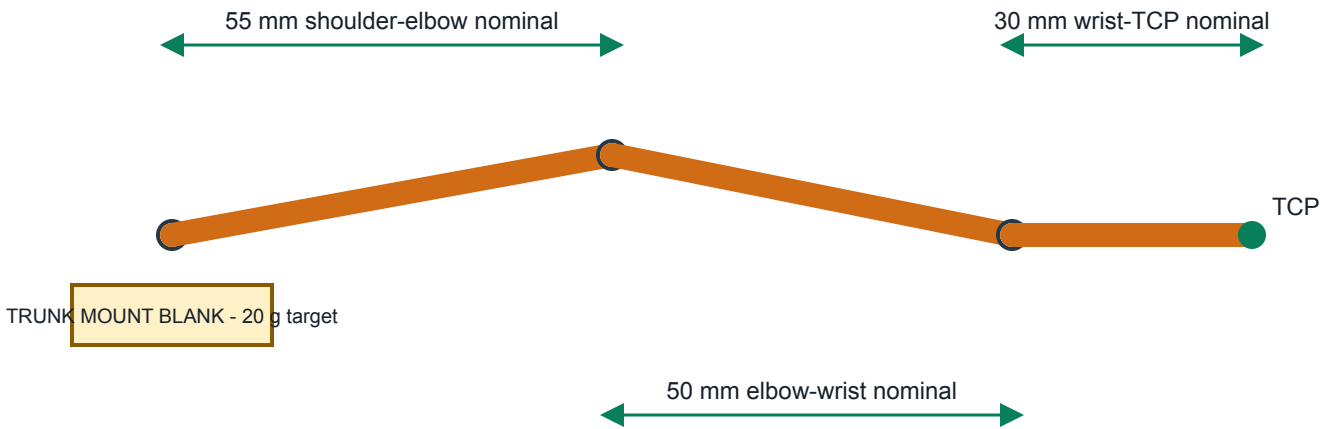


MicroDuck single arm v1-B - mechanical dimensions

ENGINEERING PROTOTYPE NOT FABRICATION RELEASE

Canonical input: hardware/microduck-arm-v1/spec/design.json | revision engineering-prototype-1 | SHA-256 57aed79ec22c...

Side elevation - nominal kinematic axis distances



MOTOR ENVELOPE AND SUPPORT-BLANK BOUNDARY

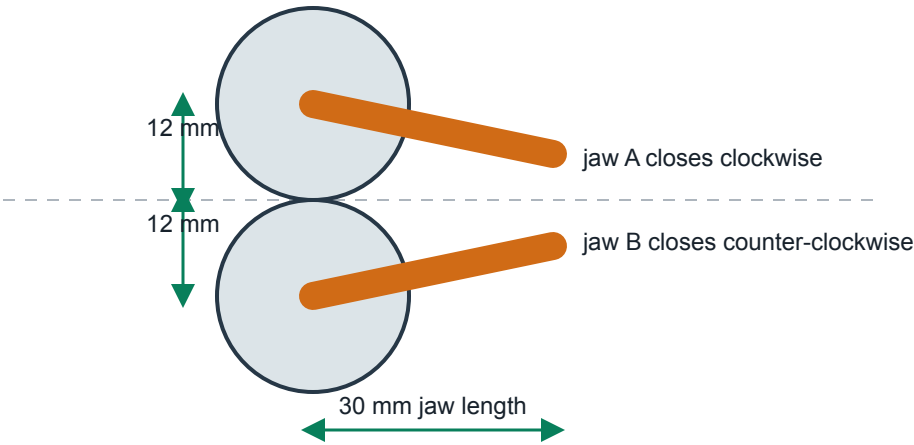
5 XL330-M288-T candidates, 18 g each; body envelope 20 x 34 x 26 mm for clearance only.

Source mesh depth is about 29 mm while manufacturer body depth is 26 mm. Do not rescale or infer a mating face.

No horn pilot, bolt circle, screw size/depth, bearing seat, cable relief, tolerance, or hole pattern is released.

Kinematic sketch omits motor boxes. Canonical local centers (mm): yaw [0,0,-25], shoulder [0,0,18], elbow [45,0,20], wrist [25,0,20], gripper [7,0,22].

Top view - counter-rotating geared gripper concept



Equal pitch gears
ratio = -1
axes parallel to +Z
centers y = +/-12 mm
rotary jaws, no sliders
left q [-0.32,0], right q [0,0.32] rad
right = -left; max closure 0.32 rad
gear teeth/module/backlash TBD

MASS AND LOAD TARGETS - UNVERIFIED

Central trunk mount target: 20 g. Complete arm cartridge target: 160 g. Motors alone: 90 g.

Required shoulder continuous capability evidence: at least 3x measured worst-case gravity torque. Evidence is missing.